

## Homework 7

程昊一

\*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

**Problem 1** (5C-1). Prove or give a counterexample: If  $T \in \mathcal{L}(V)$  and  $T^2$  has an upper-triangular matrix with respect to some basis of  $V$ , then  $T$  has an upper-triangular matrix with respect to some basis of  $V$ .

**Solution.** We give a counterexample. Let  $V = \mathbb{R}^2$  and let  $T \in \mathcal{L}(V)$  be defined by rotating every vector in  $V$  by  $90^\circ$  counterclockwise. Then  $T^2 = -I$ , which obviously has an upper-triangular matrix. However,  $T$  does not have an upper-triangular matrix with respect to any basis of  $V$  since it has no eigenvalues in  $\mathbb{R}$ .

**Problem 2** (5C-2). Suppose  $A$  and  $B$  are upper-triangular matrices of the same size, with  $\alpha_1, \dots, \alpha_n$  on the diagonal of  $A$  and  $\beta_1, \dots, \beta_n$  on the diagonal of  $B$ .

- (a) Show that  $A + B$  is an upper-triangular matrix with  $\alpha_1 + \beta_1, \dots, \alpha_n + \beta_n$  on the diagonal.
- (b) Show that  $AB$  is an upper-triangular matrix with  $\alpha_1\beta_1, \dots, \alpha_n\beta_n$  on the diagonal.

**Proof.** (a) Denote  $M_{ij}$  as the  $(i, j)$ -th entry of a matrix  $M$ . Since  $A$  and  $B$  are upper-triangular, we have  $A_{ij} = 0$  and  $B_{ij} = 0$  for all  $i > j$ . Therefore, for all  $i > j$  and  $k$ , we have

$$(A + B)_{ij} = A_{ij} + B_{ij} = 0 + 0 = 0, \quad (A + B)_{kk} = A_{kk} + B_{kk} = \alpha_k + \beta_k.$$

(b) For all  $i > j$ , we have

$$(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj} = 0,$$

since for each term in the sum, either  $i > k$  or  $k > j$ . For the diagonal entries, we have

$$(AB)_{kk} = \sum_{j=1}^n A_{kj}B_{jk} = A_{kk}B_{kk} = \alpha_k\beta_k.$$

□

**Problem 3** (5C-3). Suppose  $T \in \mathcal{L}(V)$  is invertible and  $v_1, \dots, v_n$  is a basis of  $V$  with respect to which the matrix of  $T$  is upper triangular, with  $\lambda_1, \dots, \lambda_n$  on the diagonal. Show that the matrix of  $T^{-1}$  is also upper triangular with respect to the basis  $v_1, \dots, v_n$ , with

$$\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

on the diagonal.

**Proof.** Denote  $M = \mathcal{M}(T)$  which is upper triangular. We will prove by induction on  $n = \dim V$ .

The case  $n = 1$  is trivial. Suppose that the statement holds for  $n - 1$ . We can select a basis of  $V$  with respect to which the matrix of  $T$  is upper triangular. Then we can write

$$M = \begin{pmatrix} a & X \\ 0 & A \end{pmatrix}, M^{-1} = \begin{pmatrix} b & Y \\ Z & B \end{pmatrix}, \text{ where } a, b \in \mathbb{F}, X, Y \in F^{1 \times (n-1)}, Z \in F^{(n-1) \times 1}$$

and  $A \in F^{(n-1) \times (n-1)}$  is an upper-triangular matrix.

Since  $MM^{-1} = I$ , we have

$$ab + XZ = 1, AB = I_{n-1}, aY + XB = 0, AZ = 0.$$

By the induction hypothesis,  $B = A^{-1}$  is upper triangular with diagonal entries  $\{1/\lambda : \lambda \text{ is diagonal entries of } A\}$ . Since  $A$  is invertible and  $AZ = 0$ , we have  $Z = 0$ . Then  $ab + XZ = 1$  implies that  $b = 1/a$ .

This completes the proof.  $\square$

**Problem 4** (5C-8). Suppose  $V$  is finite-dimensional,  $T \in \mathcal{L}(V)$ , and there exists a nonzero vector  $v \in V$  such that  $T^2v + 2Tv = -2v$ .

- Prove that if  $\mathbb{F} = \mathbb{R}$ , then there does not exist a basis of  $V$  with respect to which  $T$  has an upper-triangular matrix.
- Prove that if  $\mathbb{F} = \mathbb{C}$  and  $A$  is an upper-triangular matrix that equals the matrix of  $T$  with respect to some basis of  $V$ , then  $-1 + i$  or  $-1 - i$  appears on the diagonal of  $A$ .

**Proof.** (a) Denote the matrix of  $T$  with respect to some basis of  $V$  by  $A$ , such that  $A$  is upper triangular. Then the matrix of  $T^2 + 2T + 2I$  is  $A^2 + 2A + 2I$ , which is also upper triangular and with diagonal entries  $\lambda^2 + 2\lambda + 2$ , where  $\lambda$  is a diagonal entry of  $A$ . Since  $\lambda \in \mathbb{R}$ , we have  $\lambda^2 + 2\lambda + 2 > 0$ , which shows that  $A^2 + 2A + 2I$  is invertible. However, since  $T^2v + 2Tv = -2v$ , we have  $(T^2 + 2T + 2I)v = 0$ , which contradicts the invertibility of  $A^2 + 2A + 2I$ .

(b) The same argument as in (a) shows that  $A^2 + 2A + 2I$  is not invertible, which means that there exists a diagonal entry  $\lambda$  of  $A$  such that  $\lambda^2 + 2\lambda + 2 = 0$ . Solving this equation gives  $\lambda = -1 + i$  or  $\lambda = -1 - i$ .  $\square$

**Problem 5** (5C-12). Suppose  $V$  is finite-dimensional,  $T \in \mathcal{L}(V)$  has an upper-triangular matrix with respect to some basis of  $V$ , and  $U$  is a subspace of  $V$  that is invariant under  $T$ .

- (a) Prove that  $T|_U$  has an upper-triangular matrix with respect to some basis of  $U$ .
- (b) Prove that the quotient operator  $T/U$  has an upper-triangular matrix with respect to some basis of  $V/U$ .

**Proof.** (a) Let  $v_1, \dots, v_n$  be a basis of  $V$  such that the matrix of  $T$  with respect to this basis is upper triangular, and let  $V_k = \text{span}\{v_1, \dots, v_k\}$ . Then,  $\dim V_k = k$ , and  $\{0\} = V_0 \subset V_1 \subset \dots \subset V_n = V$  is a sequence of subspaces that are invariant under  $T$ .

We will show that  $\{0\} = (V_0 \cap U) \subset (V_1 \cap U) \subset \dots \subset (V_n \cap U) = U$  is a sequence of subspaces that are invariant under  $T|_U$  with  $\dim(V_{k+1} \cap U) - \dim(V_k \cap U) \leq 1$  for all  $k = 0, \dots, n-1$ . Since  $V_{k+1} = V_k + \text{span}\{v_{k+1}\}$ , we have

$$V_{k+1} \cap U = (V_k + \text{span}\{v_{k+1}\}) \cap U = (V_k \cap U) + (\text{span}\{v_{k+1}\} \cap U).$$

Therefore

$$\dim(V_{k+1} \cap U) = \dim(V_k \cap U) + \dim(\text{span}\{v_{k+1}\} \cap U) - \dim(V_k \cap \text{span}\{v_{k+1}\} \cap U) \leq \dim(V_k \cap U) + 1.$$

Denote  $m = \dim U$ . Then there exists  $0 = i_0 < i_1 < \dots < i_m = n$  such that  $\dim(V_{i_k} \cap U) = k$  for all  $k = 0, \dots, m$ . Pick  $u_k \in (V_{i_k} \cap U) - (V_{i_{k-1}} \cap U)$ , then  $\{u_1, \dots, u_m\}$  is a basis of  $U$ . Moreover, for any  $1 \leq k \leq m$ ,  $\{u_1, \dots, u_k\}$  is a basis of  $V_{i_k} \cap U$ . Therefore, the matrix of  $T|_U$  with respect to the basis  $u_1, \dots, u_m$  is upper triangular.

(b) We will consider the sequence of subspaces

$$\{0\} = (V_0 + U)/U \subset (V_1 + U)/U \subset \dots \subset (V_n + U)/U = V/U.$$

Note that

$$\begin{aligned} \dim((V_{k+1} + U)/U) &= \dim(V_{k+1} + U) - \dim U = \dim V_{k+1} + \dim U - \dim(V_{k+1} \cap U) - \dim U \\ &= \dim V_{k+1} - \dim(V_{k+1} \cap U) \leq 1 + \dim V_k - \dim(V_k \cap U) \\ &= 1 + \dim((V_k + U)/U). \end{aligned}$$

Then the similar argument as in (a) shows that there exists a basis of  $V/U$  with respect to which the matrix of  $T/U$  is upper triangular.  $\square$

**Problem 6 (5C-13).** Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ . Suppose there exists a subspace  $U$  of  $V$  that is invariant under  $T$  such that  $T|_U$  has an upper-triangular matrix with respect to some basis of  $U$  and also  $T/U$  has an upper-triangular matrix with respect to some basis of  $V/U$ . Prove that  $T$  has an upper-triangular matrix with respect to some basis of  $V$ .

**Proof.** Denote  $\{u_1, \dots, u_m\}$  the basis of  $U$  such that the matrix of  $T|_U$  with respect to this basis is upper triangular, and denote  $\{v_1 + U, \dots, v_n + U\}$  the basis of  $V/U$  such that the matrix of  $T/U$  with respect to this basis is upper triangular. We will show that  $\{u_1, \dots, u_m, v_1, \dots, v_n\}$  is a basis of  $V$  such that the matrix of  $T$  with respect to this basis is upper triangular.

First, for any  $1 \leq k \leq m$ , we have  $\text{span}\{u_1, \dots, u_k\}$  is  $T$ -invariant since the matrix of  $T|_U$  with respect to the basis  $u_1, \dots, u_m$  is upper triangular. Next, for any  $1 \leq k \leq n$ , we have

$$\begin{aligned} T(v_k) + U &= (T/U)(v_k + U) \in \text{span}\{v_1 + U, \dots, v_k + U\} \subset \text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\} + U \\ \implies T(v_k) &\in \text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\}. \end{aligned}$$

Then,  $\text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\}$  is  $T$ -invariant for all  $1 \leq k \leq n$ . Therefore, the matrix of  $T$  with respect to the basis  $u_1, \dots, u_m, v_1, \dots, v_n$  is upper triangular.  $\square$

**Problem 7** (5C-14). Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ . Prove that  $T$  has an upper-triangular matrix with respect to some basis of  $V$  if and only if the dual operator  $T^*$  has an upper-triangular matrix with respect to some basis of the dual space  $V^*$ .

**Proof.** Since  $V$  is finite-dimensional, we can treat  $V^{**}$  as  $V$ . Then we only need to show “ $\implies$ ”.

Let  $\mathcal{B} := \{v_1, \dots, v_n\}$  be a basis of  $V$  such that the matrix of  $T$  with respect to this basis is upper triangular, and let  $\mathcal{B}^* := \{\varphi_1, \dots, \varphi_n\}$  be the dual basis of  $V^*$ . Then we have

$$\mathcal{M}(T, \mathcal{B}) = (\mathcal{M}(T^*, \mathcal{B}^*))^T.$$

Then the matrix of  $T^*$  with respect to the basis  $\mathcal{B}^*$  is lower-triangular matrix, which implies that the matrix of  $T^*$  with respect to the basis  $\{\varphi_n, \varphi_{n-1}, \dots, \varphi_1\}$  is upper triangular.  $\square$

**Problem 8** (5D-1). Suppose  $V$  is a finite-dimensional complex vector space and  $T \in \mathcal{L}(V)$ .

- (a) Prove that if  $T^4 = I$ , then  $T$  is diagonalizable.
- (b) Prove that if  $T^4 = T$ , then  $T$  is diagonalizable.
- (c) Give an example of an operator  $T \in \mathcal{L}(\mathbb{C}^2)$  such that  $T^4 = T^2$  and  $T$  is not diagonalizable.

**Proof.** (a) We have

$$(T - I)(T + I)(T - iI)(T + iI) = 0.$$

Since the four polynomials  $z - 1, z + 1, z - i, z + i$  are pairwise relatively prime, we have  $V = E(1, T) \oplus E(-1, T) \oplus E(i, T) \oplus E(-i, T)$ . Therefore,  $T$  is diagonalizable.

(b) Similar to (a), we have

$$V = E(0, T) \oplus E(1, T) \oplus E(\omega, T) \oplus E(\omega^2, T), \text{ where } \omega = e^{2\pi i/3}.$$

Therefore,  $T$  is diagonalizable.

(c) Let  $T \in \mathcal{L}(\mathbb{C}^2)$  be defined by  $T(z, w) = (0, z)$  for all  $(z, w) \in \mathbb{C}^2$ . Then  $T^2(z, w) = T(0, z) = (0, 0)$ , so  $T^2 = 0$ , but  $T$  is not diagonalizable since  $T$  has only one eigenvalue 0 but the eigenspace  $E(0, T)$  is one-dimensional.  $\square$

**Problem 9** (5D-3). Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ . Prove that if the operator  $T$  is diagonalizable, then

$$V = \ker T \oplus \operatorname{im} T.$$

**Proof.** Let  $\{v_1, v_2, \dots, v_n\}$  be all the nonzero eigenvectors of  $T$  such that  $Tv_k = \lambda_k v_k$  for all  $k = 1, \dots, n$ . Then  $\{v_1, v_2, \dots, v_n\}$  is a basis of  $V$ . It's clear that

$$\ker T = \operatorname{span}\{v_i : \lambda_i = 0\} \text{ and } \operatorname{im} T = \operatorname{span}\{v_i : \lambda_i \neq 0\}.$$

Then  $V = \ker T \oplus \operatorname{im} T$ . □

**Problem 10** (5D-5). Suppose  $V$  is a finite-dimensional complex vector space and  $T \in \mathcal{L}(V)$ . Prove that  $T$  is diagonalizable if and only if

$$V = \ker(T - \lambda I) \oplus \operatorname{im}(T - \lambda I)$$

for every  $\lambda \in \mathbb{C}$ .

**Proof.** “ $\implies$ ”: Let  $\{\lambda_i\}$  be all the eigenvalues of  $T$  and  $\{v_i\}$  be the corresponding eigenvectors. Then  $\{v_i\}$  is a basis of  $V$ . For any  $\lambda \in \mathbb{C}$ , similar to the previous problem, we have

$$\ker(T - \lambda I) = \operatorname{span}\{v_i : \lambda_i = \lambda\} \text{ and } \operatorname{im}(T - \lambda I) = \operatorname{span}\{v_i : \lambda_i \neq \lambda\}.$$

Therefore,  $V = \ker(T - \lambda I) \oplus \operatorname{im}(T - \lambda I)$ .

“ $\impliedby$ ”: We will prove first that

$$\ker(T - \lambda I)^k = \ker(T - \lambda I) \text{ for any } k \geq 1.$$

We will use induction on  $k$ . The case  $k = 1$  is trivial. Assume that the statement holds for some  $k \geq 1$ . For any  $v \in \ker(T - \lambda I)^{k+1}$ , we have  $(T - \lambda I)v \in \ker(T - \lambda I)^k = \ker(T - \lambda I)$ . Therefore,  $(T - \lambda I)v \in \ker(T - \lambda I) \cap \operatorname{im}(T - \lambda I) = \{0\}$ , which implies that  $v \in \ker(T - \lambda I)$ . Therefore,  $\ker(T - \lambda I)^{k+1} = \ker(T - \lambda I)$ .

Denote all distinct eigenvalues of  $T$  by  $\lambda_1, \dots, \lambda_m$ . Denote the minimal polynomial by  $m(\lambda) = (\lambda - \lambda_1)^{d_1} \dots (\lambda - \lambda_m)^{d_m}$ . Then

$$V = \bigoplus_{k=1}^m \ker(T - \lambda_k I)^{d_k} = \bigoplus_{k=1}^m \ker(T - \lambda_k I).$$

Therefore,  $T$  is diagonalizable. □

**Problem 11** (5D-11). Find  $T \in \mathcal{L}(\mathbb{C}^3)$  such that 6 and 7 are eigenvalues of  $T$  and such that  $T$  does not have a diagonal matrix with respect to any basis of  $\mathbb{C}^3$ .

**Solution.** Take  $T(z_1, z_2, z_3) = (7z_1, 6z_2 + z_3, 6z_3)$ . Then  $T$  has eigenvalues 6 and 7, but  $T$  is not diagonalizable since the eigenspace  $E(6, T)$  is one-dimensional.

**Problem 12** (5D-16). Suppose that  $T \in \mathcal{L}(V)$  is diagonalizable. Let  $\lambda_1, \dots, \lambda_m$  denote the distinct eigenvalues of  $T$ . Prove that a subspace  $U$  of  $V$  is invariant under  $T$  if and only if there exist subspaces  $U_1, \dots, U_m$  of  $V$  such that  $U_k \subseteq E(\lambda_k, T)$  for each  $k$  and

$$U = U_1 \oplus \cdots \oplus U_m.$$

**Proof.** “ $\Leftarrow$ ”: Since  $E(\lambda_k, T)$  is invariant under  $T$  for each  $k$ , the subspace  $U_k \subseteq E(\lambda_k, T)$  is also invariant under  $T$ . Therefore,  $U = U_1 \oplus \cdots \oplus U_m$  is invariant under  $T$ .

“ $\Rightarrow$ ”: Let  $U_k = U \cap E(\lambda_k, T)$ , we will show that  $U = U_1 \oplus \cdots \oplus U_m$ . Since  $\bigoplus_{k=1}^m E(\lambda_k, T) = V$  is direct sum, we have  $\bigoplus_{k=1}^m U_k$  is also direct sum. For any  $u \in U$ , we can write  $u = u_1 + \cdots + u_m$  with  $u_k \in E(\lambda_k, T)$  for all  $k = 1, \dots, m$ . Since  $U$  is invariant under  $T$ , we have

$$u_k = \prod_{i \neq k} \frac{(T - \lambda_i I)}{\lambda_k - \lambda_i}(u) \in U \implies u_k \in U \cap E(\lambda_k, T) = U_k$$

for all  $k = 1, \dots, m$ . Therefore,  $U_1 \oplus \cdots \oplus U_m \subset U$ . On the other hand, for any  $u_k \in U_k$ , we have  $u_k \in U$  and  $u_1 + \cdots + u_k \in U$ . Therefore,  $U = U_1 \oplus \cdots \oplus U_m$ .  $\square$

**Problem 13** (5D-18). Suppose that  $T \in \mathcal{L}(V)$  is diagonalizable and  $U$  is a subspace of  $V$  that is invariant under  $T$ . Prove that the quotient operator  $T/U$  is a diagonalizable operator on  $V/U$ .

**Proof.** Denote the distinct eigenvectors of  $T$  by  $v_1, \dots, v_n$  with  $Tv_k = \lambda_k v_k$ . Then we have

$$(T/U)(v_k + U) = Tv_k + U = \lambda_k v_k + U = \lambda_k(v_k + U)$$

for all  $k = 1, \dots, n$ . Then  $v_k + U$  is an eigenvector of  $T/U$  corresponding to the eigenvalue  $\lambda_k$ .

Note that  $\text{span}\{v_k + U : 1 \leq k \leq n\} = \text{span}\{v_k : 1 \leq k \leq n\} + U = V/U$ , so we can pick a subset of  $\{v_k + U : 1 \leq k \leq n\}$  as a basis of  $V/U$ . Therefore,  $T/U$  is diagonalizable.  $\square$

**Problem 14** (5D-19). Prove or give a counterexample: If  $T \in \mathcal{L}(V)$  and there exists a subspace  $U$  of  $V$  that is invariant under  $T$  such that  $T|_U$  and  $T/U$  are both diagonalizable, then  $T$  is diagonalizable.

**Proof.** We give a counterexample. Let  $T \in \mathcal{L}(\mathbb{C}^2)$  be defined by  $T(z, w) = (z + w, w)$  for all  $(z, w) \in \mathbb{C}^2$ . Then  $T|_{\text{span}\{(0,1)\}}$  is diagonalizable and  $T/\text{span}\{(0,1)\}$  is also diagonalizable, but  $T$  is not diagonalizable since the eigenspace  $E(1, T)$  is one-dimensional.  $\square$

**Problem 15** (5E-4). Prove or give a counterexample: If  $A$  is a diagonal matrix and  $B$  is an upper-triangular matrix of the same size as  $A$ , then  $A$  and  $B$  commute.

**Solution.** We give a counterexample. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = BA.$$

**Problem 16** (5E-7). Suppose  $V$  is a complex vector space,  $S \in \mathcal{L}(V)$  is diagonalizable, and  $T \in \mathcal{L}(V)$  commutes with  $S$ . Prove that there is a basis of  $V$  such that  $S$  has a diagonal matrix with respect to this basis and  $T$  has an upper-triangular matrix with respect to this basis.

**Proof.** Let  $\{\lambda_i\}$  be all the distinct eigenvalues of  $S$ . Then for any  $\lambda_k$  and any  $v \in E(\lambda_k, S)$ , we have

$$S(Tv) = T(Sv) = T(\lambda_k v) = \lambda_k(Tv),$$

which implies that  $Tv \in E(\lambda_k, S)$ . Therefore,  $E(\lambda_k, S)$  is invariant under  $T$  for all  $k$ . Then we can pick a basis of  $E(\lambda_k, S)$  with respect to which the matrix of  $T|_{E(\lambda_k, S)}$  is upper triangular for all  $k$ .

Then the union of these bases is a basis of  $V$ , with respect to which the matrix of  $S$  is diagonal and the matrix of  $T$  is upper triangular. Moreover, since  $V = \bigoplus_{k=1}^n E(\lambda_k, S)$ , we can infer that  $\mathcal{M}(T)$  is block diagonal with blocks  $\mathcal{M}(T|_{E(\lambda_k, S)})$  for  $k = 1, \dots, n$ , which is upper triangular.  $\square$

**Problem 17** (5E-9). Suppose  $V$  is a finite-dimensional nonzero complex vector space. Suppose that  $\mathcal{E} \subseteq \mathcal{L}(V)$  is such that  $S$  and  $T$  commute for all  $S, T \in \mathcal{E}$ .

- (a) Prove that there is a vector in  $V$  that is an eigenvector for every element of  $\mathcal{E}$ .
- (b) Prove that there is a basis of  $V$  with respect to which every element of  $\mathcal{E}$  has an upper-triangular matrix.

**Proof.** (a) We will use induction on  $\dim V$ . The case  $\dim V = 1$  is trivial. Assume that the statement holds for any complex vector space with dimension less than  $n$  for some  $n \geq 2$ .

Let  $S \in \mathcal{E}$  be a non-scalar operator, then there exists an eigenvalue  $\lambda$  of  $S$  such that the eigenspace  $E(\lambda, S)$  is a nonzero proper subspace of  $V$ . Since  $S$  commutes with every element of  $\mathcal{E}$ , we can infer that  $E(\lambda, S)$  is  $T$ -invariant for all  $T \in \mathcal{E}$ .

Then we can apply the induction hypothesis to  $E(\lambda, S)$  and infer that there exists a vector in  $E(\lambda, S)$  that is an eigenvector for every element of  $\mathcal{E}$ .

(b) We will use induction on  $\dim V$ . The case  $\dim V = 1$  is trivial. Assume that the statement holds for any complex vector space with dimension less than  $n$  for some  $n \geq 2$ .

From (a), we can pick a vector  $v$  that is an eigenvector for every element of  $\mathcal{E}$ . Denote  $U = \text{span}\{v\}$ , and select a subspace  $W$  of  $V$  such that  $V = U \oplus W$ . Denote  $P$  as the projection from  $V$  to  $W$  along  $U$ . For any  $T \in \mathcal{E}$ , define  $f(T) \in \mathcal{L}(W)$  by  $f(T)(w) = P(Tw)$  for all  $w \in W$ . Then for any  $S, T \in \mathcal{E}$ , we have

$$f(S)f(T)(w) = P(SP(Tw)) = P(STw) = P(TSw) = P(TP(Sw)) = f(T)f(S)(w).$$

Therefore,  $f(S)$  and  $f(T)$  commute for all  $S, T \in \mathcal{E}$ . Since  $\dim W < \dim V$ , we can apply the induction hypothesis to  $W$  and infer that there exists a basis  $\mathcal{B}'$  of  $W$  with respect to which every element of  $f(\mathcal{E})$  has an upper-triangular matrix.

Consider the basis  $\mathcal{B} = \{v\} \cup \mathcal{B}'$  of  $V$ . Then it's easy to check that for all  $T \in \mathcal{E}$ , we have

$$\mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} \lambda & * \\ 0 & \mathcal{M}(f(T), \mathcal{B}') \end{pmatrix},$$

where  $Tv = \lambda v$  for all  $v \in U$ . Since  $\mathcal{M}(f(T), \mathcal{B}')$  is upper triangular, we have  $\mathcal{M}(T, \mathcal{B})$  is also upper triangular.  $\square$